

### 1.1.3 Multi-Tasking Operating Systems

All modern OSes—Windows, Linux, Unix, Mac OS X and others—are capable of multi-tasking: running many programs at the same time. Resources are shared by all the programs turn by turn on a ‘time-sharing’ basis. The operating system decides which program is allowed to use the computer’s resources and when.

Multi-tasking operating systems are loved by one and all—users can run many different applications at the same time, programmers can write programs that perform multiple tasks simultaneously rather than waiting for one task to finish before beginning the other, and finally, chip designers are happy because multi-tasking makes better use of the capabilities of their chips. We will talk more about multi-tasking later in this chapter.

### 1.1.4 Multi-User Operating Systems

Multi-User Operating Systems are designed so that many users can use the system’s resources at the same time. Such OSes are called “Server” or “Mainframe” operating systems. In addition to managing different tasks, multi-user OSes need to ensure that each user is getting the most out of the machine, and that problems with one user do not affect other users.

This was usually the scenario in older corporate environments where all resources would be present on one central computer, and all users would connect to it from their terminals—basically a keyboard and a monitor—to access their programs.

### 1.1.5 Multi-Processing Operating Systems

To do their job better, many OSes support the use of more than one processor. This way, tasks can be carried out in parallel, speeding up the system. This was especially useful for servers, which need to perform a large number of operations as quickly as possible.

There are two ways for an operating system to work in a multi-processor setup. The first, called *Asymmetric Multi-processing*, gives one processor to the OS to use exclusively for its own tasks. Tasks